ROC Curve and its Usage in Machine Learning

Introduction

The Receiver Operating Characteristic (ROC) Curve is a graphical tool used to evaluate the performance of a binary classifier. It helps visualize the trade-off between sensitivity (true positive rate) and specificity (1 – false positive rate).

How it Works

- **True Positive Rate (TPR)**:

$TPR = TP/TP+FN$

- **False Positive Rate (FPR)**:

$FPR = FP/FP+TN$

- The ROC curve plots TPR vs FPR at different thresholds.

Area Under Curve (AUC)

- **AUC = 1**: Perfect classifier.
- **AUC = 0.5**: Random guessing.
- **AUC < 0.5**: Worse than random.

Applications

- **Medical Diagnosis**: ROC curves help evaluate tests that detect diseases.
- **Credit Scoring**: Assessing the ability of a model to predict loan defaults.
- **Spam Detection**: Balancing false positives (legit email marked spam) and false negatives (spam missed).
- **Model Comparison**: Choosing the best classifier among several models.

Advantages

- Independent of class distribution.
- Provides a full picture of model performance across thresholds.

Limitations

- Doesn’t consider cost of misclassification.
- For imbalanced datasets, **Precision-Recall Curve** may be more informative.